

Integration and Poincaré Duality on Orbifolds

Conventions and the central distinction

Let $\mathcal{X}$ be a smooth complex orbifold of complex dimension n and real dimension $d = 2n$, without boundary. Write

$$X = |\mathcal{X}|$$

for its underlying, or coarse, topological space. We assume the usual Hausdorff and paracompactness conditions. We allow **ineffective** finite group actions, as in Zvonkine's notes [Z]. Unless stated otherwise, the cohomology under discussion is ordinary cohomology, not Chen–Ruan cohomology.

The same constructions apply to oriented smooth real orbifolds, provided all chart group actions preserve orientation.

The basic distinction is:

The rational cohomology ring is canonically the cohomology ring of the coarse space, but integration uses an orbifold-normalized fundamental class.

There are three related but different objects:

1. the ordinary topological fundamental class of the coarse space;
2. the stabilizer-weighted fundamental class of the orbifold or of a suborbifold;
3. the cohomological Poincaré dual of such a weighted cycle.

In particular, the coefficient in a weighted **homology** cycle need not be the coefficient in its **cohomological** Poincaré dual. The latter depends on the normalization of the ambient fundamental class as well.

The starting points in [Z] are §1.3, Theorem 1.20, and Remark 1.21. The discussion below supplies the differential-form constructions, the precise rational duality statement, and the normalization conventions.

1. Differential forms on an orbifold

1.1. Local definition and compatibility

An orbifold chart has the form

$$[U/G] \longrightarrow \mathcal{X}, \quad q : U \longrightarrow V \subset X,$$

where U is an open subset of $\mathbb{C}^n$ and G is finite.

A **smooth differential k -form** on this chart is a form

$$\alpha_U \in \Omega^k(U)$$

satisfying

$$g^* \alpha_U = \alpha_U \quad (g \in G).$$

A global form consists of such invariant forms on all charts, compatible under chart embeddings: if

$\iota : U' \hookrightarrow U$ is a chart embedding, then

$$\iota^* \alpha_U = \alpha_{U'}.$$

This gives a sheaf $\Omega_{\mathcal{X}}^k$ on X , and a differential graded algebra

$$(\Omega^\bullet(\mathcal{X}), d, \wedge).$$

Exterior differentiation and wedge product are defined upstairs. They preserve invariance and compatibility because pullback commutes with both operations. This is the expanded form of the differential-form definition in [Z, §1.3].

The forms here are **smooth real forms**, or their complexifications, not merely holomorphic forms. On a complex orbifold, one also has the (p, q) -decomposition and the operators ∂ and $\bar{\partial}$. Integration concerns real degree $2n$; a complex-valued form of that degree has type (n, n) .

1.2. Orbifold forms versus smooth forms on the coarse space

Even when the coarse space happens to be a manifold, its ordinary smooth structure need not identify ordinary smooth forms with orbifold forms.

For example, consider

$$[\mathbb{C}/\mu_m], \quad z \longmapsto \zeta z.$$

The function $|z|^2$ is a smooth invariant function upstairs, hence a smooth orbifold function. In the coarse coordinate $w = z^m$, it becomes

$$|w|^{2/m},$$

which is not smooth at $w = 0$ when $m > 1$.

Thus differential forms should be defined on the orbifold charts, rather than by choosing an ordinary smooth structure on the coarse space.

2. The de Rham theorem and rational cohomology

2.1. Orbifold de Rham theorem

Theorem. There are natural isomorphisms

$$H_{\text{dR}}^k(\mathcal{X}) \cong H^k(X; \mathbb{R}),$$

compatible with products. Compactly supported orbifold forms likewise compute

$$H_{\text{dR},c}^k(\mathcal{X}) \cong H_c^k(X; \mathbb{R}).$$

These statements are also part of the differential-geometric cohomology formalism for Deligne–Mumford stacks in [B].

Proof. There are two ingredients.

First, the ordinary Poincaré lemma can be made equivariant by averaging. Near a point, shrink to a chart whose upstairs domain is a ball. If

$$\alpha \in \Omega^k(U)^G, \quad d\alpha = 0, \quad k > 0,$$

then $\alpha = d\beta$ for some ordinary form β . Although β need not be invariant, the form

$$\beta_{\text{av}} = \frac{1}{|G|} \sum_{g \in G} g^* \beta$$

is invariant and satisfies

$$d\beta_{\text{av}} = \alpha.$$

Consequently, the sheaf sequence

$$0 \longrightarrow \mathbb{R}_X \longrightarrow \Omega_{\mathcal{X}}^0 \xrightarrow{d} \Omega_{\mathcal{X}}^1 \xrightarrow{d} \cdots$$

is exact.

Second, smooth orbifold partitions of unity exist. One constructs cutoffs upstairs, averages them under the finite chart groups, and normalizes their sum. Therefore the sheaves $\Omega_{\mathcal{X}}^k$ are fine, and this complex is a fine resolution of $\mathbb{R}_X$. Taking global sections, or compactly supported sections, proves the assertions. $\square$

2.2. What “over $\mathbb{Q}$ ” means

Following the coarse-space convention in [Z], set

$$H^k(\mathcal{X}; \mathbb{Q}) := H^k(X; \mathbb{Q}).$$

For compact $\mathcal{X}$, or more generally in the finite-type situation,

$$H_{\text{dR}}^k(\mathcal{X}) \cong H^k(X; \mathbb{Q}) \otimes_{\mathbb{Q}} \mathbb{R}.$$

A **rational de Rham class** is a real de Rham class lying in the image of $H^k(X; \mathbb{Q})$. It does not mean that the coefficients of a differential form are pointwise rational.

There is also an intrinsic cohomology defined using a classifying space $B\mathcal{X}$ of the orbifold. The natural map to the coarse space induces a canonical rational isomorphism

$$H^k(X; \mathbb{Q}) \xrightarrow{\sim} H^k(B\mathcal{X}; \mathbb{Q}).$$

Locally, the reason is that for a finite group G ,

$$H^{>0}(BG; \mathbb{Q}) = 0 :$$

averaging makes the invariants functor exact. The coarse-space comparison is stated in [B, Proposition 36]. Thus the coarse-space convention agrees with intrinsic orbifold cohomology over $\mathbb{Q}$. The analogous integral comparison need not hold.

3. Integration of top-degree differential forms

3.1. Definition

Let

$$\omega \in \Omega_c^d(\mathcal{X}).$$

Choose charts $[U_i/G_i]$ and a subordinate smooth orbifold partition of unity $\{\rho_i\}$. Let ω_i and $\tilde{\rho}_i$ be their representatives upstairs. Define

$$\int_{\mathcal{X}} \omega = \sum_i \frac{1}{|G_i|} \int_{U_i} \tilde{\rho}_i \omega_i. \quad (3.1)$$

The support conditions make each ordinary integral meaningful, and local finiteness makes the sum meaningful.

The denominator is the order of the full chart group, including its ineffective kernel. This is the global form of Zvonkine’s chartwise integration rule [Z, §1.3].

3.2. Independence of choices

Suppose a sufficiently small genuine open subchart is $[U'/H] \subset [U/G]$. The inverse image of its coarse footprint in U consists of $[G:H]$ translates of U' . For an invariant form supported over that footprint,

$$\frac{1}{|G|} \int_U \omega = \frac{[G:H]}{|G|} \int_{U'} \omega = \frac{1}{|H|} \int_{U'} \omega.$$

This proves compatibility with chart refinement. Passing to a common refinement of two chart covers and two partitions of unity proves that (3.1) is independent of all choices.

Here “open subchart” means that the orbifold structure, including all stabilizers, is retained; it is not an arbitrary map $[U'/H] \rightarrow [U/G]$.

3.3. Basic normalizations

For a finite global quotient,

$$\int_{[M/G]} \omega = \frac{1}{|G|} \int_M \tilde{\omega}. \quad (3.2)$$

If the action is free, this is exactly ordinary integration on the manifold M/G , since $M \rightarrow M/G$ is a covering of degree $|G|$.

If the action is trivial, the same formula still applies:

$$\int_{M \times BG} \omega = \frac{1}{|G|} \int_M \omega.$$

In particular,

$$\boxed{\int_{BG} 1 = \frac{1}{|G|}.} \quad (3.3)$$

This zero-dimensional example fixes the stabilizer convention for all later constructions.

3.4. Stokes' theorem

Proposition. If $\eta \in \Omega_c^{d-1}(\mathcal{X})$, then

$$\int_{\mathcal{X}} d\eta = 0.$$

Proof. Apply ordinary Stokes' theorem upstairs to the compactly supported forms $\tilde{\rho}_i \eta_i$:

$$\int_{U_i} d(\tilde{\rho}_i \eta_i) = 0.$$

Expand the derivative and sum with weights $1/|G_i|$. The terms involving $d\rho_i$ cancel, after passage to common charts, because

$$\sum_i d\rho_i = 0.$$

The remaining sum is $\int_{\mathcal{X}} d\eta$. $\square$

Hence integration descends to a functional

$$\int_{\mathcal{X}} : H_c^d(X; \mathbb{R}) \longrightarrow \mathbb{R}.$$

For compact $\mathcal{X}$, this is a functional on ordinary top-degree cohomology.

The integral of an arbitrary smooth top-degree form is a real number, not necessarily rational.

Rationality arises when its cohomology class is rational.

4. Fundamental classes and rational Poincaré duality

4.1. The coarse space is a rational homology manifold

Proposition. For every $x \in X$,

$$H_j(X, X \setminus \{x\}; \mathbb{Q}) \cong \begin{cases} \mathbb{Q}, & j = d, \\ 0, & j \neq d. \end{cases} \quad (4.1)$$

The complex orientation gives a compatible choice of generators of the nonzero local homology groups.

Local explanation. A finite group action can be linearized near a fixed point, so a sufficiently small neighborhood has the form B^d/G_x , with link S^{d-1}/G_x . For a finite group action of this kind,

$$H^\bullet(S^{d-1}/G_x; \mathbb{Q}) \cong H^\bullet(S^{d-1}; \mathbb{Q})^{G_x}.$$

This follows, for example, from an equivariant triangulation and averaging; freeness is not required. The action preserves orientation, automatically in the complex case. Thus the quotient link has the rational homology of a sphere, which gives (4.1). In dimension zero, (4.1) is immediate.

Consequently, a connected compact X has an orientation fundamental class

$$[X]_{\text{top}} \in H_d(X; \mathbb{Q}).$$

It is normalized by the ordinary orientation on the dense smooth locus of the coarse space. Although X may be singular, it satisfies Poincaré duality with rational coefficients as an oriented rational homology manifold.

4.2. The orbifold fundamental class

Assume $\mathcal{X}$ is connected, and let

$$r_{\mathcal{X}} = |\text{stabilizer of a generic point of } \mathcal{X}|.$$

Define

$$\boxed{[\mathcal{X}]_{\text{orb}} = \frac{1}{r_{\mathcal{X}}} [X]_{\text{top}} \in H_d(X; \mathbb{Q}).} \quad (4.2)$$

For a disconnected orbifold, make this definition componentwise. This is the convention of [Z, Remark 1.21] applied to the whole orbifold.

Proposition. Under the de Rham isomorphism,

$$\boxed{\int_{\mathcal{X}} \omega = \langle [\omega]_{\text{dR}}, [\mathcal{X}]_{\text{orb}} \rangle.} \quad (4.3)$$

Explanation of the factor. Over the principal stratum, the orbifold is locally a manifold with an ineffective stabilizer of order $r_{\mathcal{X}}$. Thus its local integral is $1/r_{\mathcal{X}}$ times ordinary integration. The exceptional strata have positive codimension and contribute no additional terms to integration of a smooth top-degree form. Equivalently, one can determine the top-degree functional on a form supported in a small ball in the principal stratum.

For an effective connected orbifold,

$$r_{\mathcal{X}} = 1, \quad [\mathcal{X}]_{\text{orb}} = [X]_{\text{top}}.$$

Special points with larger stabilizers do **not** impose a further global division of the top-dimensional fundamental class. Their stabilizers matter when one integrates on cycles supported on those strata.

For a connected global quotient $\mathcal{X} = [M/G]$, one may also write

$$[\mathcal{X}]_{\text{orb}} = \frac{1}{|G|} q_* [M]_{\text{top}},$$

because the generic degree of the coarse quotient map $q : M \rightarrow M/G$ is $|G|/r_{\mathcal{X}}$.

4.3. The precise rational duality theorem

Theorem — Rational Poincaré duality. Let $\mathcal{X}$ be a compact oriented smooth orbifold of real dimension d , without boundary. Ineffective stabilizers are allowed. Then cap product with the orbifold fundamental class gives an isomorphism

$$D_{\mathcal{X}} : H^k(X; \mathbb{Q}) \xrightarrow{\sim} H_{d-k}(X; \mathbb{Q}), \quad a \longmapsto a \cap [\mathcal{X}]_{\text{orb}}. \quad (4.4)$$

Equivalently, the pairing

$$H^k(X; \mathbb{Q}) \times H^{d-k}(X; \mathbb{Q}) \longrightarrow \mathbb{Q}, \quad (a, b) \longmapsto \int_{\mathcal{X}} a \smile b \quad (4.5)$$

is perfect. This is the orbifold-normalized form of [Z, Theorem 1.20].

We use the cap-product convention characterized by

$$\langle b, D_{\mathcal{X}}(a) \rangle = \int_{\mathcal{X}} a \smile b.$$

Thus, for every class

$$c \in H_{d-k}(X; \mathbb{Q}),$$

there is a unique class

$$\mathrm{PD}_{\mathcal{X}}(c) \in H^k(X; \mathbb{Q})$$

such that

$$\boxed{\int_{\mathcal{X}} \mathrm{PD}_{\mathcal{X}}(c) \smile b = \langle b, c \rangle \quad \text{for all } b \in H^{d-k}(X; \mathbb{Q}).} \quad (4.6)$$

Proof. By (4.1), cap product with $[X]_{\mathrm{top}}$ gives rational Poincaré duality. Replacing this class by the nonzero rational multiple $[X]_{\mathrm{top}}/r_{\mathcal{X}}$ preserves invertibility. Equation (4.3) identifies the resulting pairing with differential-form integration. The argument is applied separately on each connected component. $\square$

This proves duality over $\mathbb{Q}$, not just over $\mathbb{R}$: the construction uses rational local homology and a rational fundamental class. In particular,

$$\int_{\mathcal{X}} a \in \mathbb{Q} \quad \text{for } a \in H^d(X; \mathbb{Q}).$$

4.4. Noncompact orbifolds

For a noncompact oriented orbifold, the fundamental class lies in Borel–Moore homology, and the corresponding duality statements are

$$\boxed{H_c^k(X; \mathbb{Q}) \cong H_{d-k}(X; \mathbb{Q}), \quad H^k(X; \mathbb{Q}) \cong H_{d-k}^{\mathrm{BM}}(X; \mathbb{Q}).} \quad (4.7)$$

Here Borel–Moore homology means homology represented by locally finite chains. The same generic-stabilizer normalization is used for the Borel–Moore fundamental class.

In the finite-type situation, integration gives a perfect pairing

$$H^k(X; \mathbb{Q}) \times H_c^{d-k}(X; \mathbb{Q}) \longrightarrow \mathbb{Q}.$$

Thus on an open moduli space one must retain compact supports on one side. The compact theorem applies directly to a smooth orbifold compactification.

5. Poincaré duals of suborbifolds: the stabilizer ratio

Let

$$i : \mathcal{Y} \hookrightarrow \mathcal{X}$$

be a closed smooth complex suborbifold of complex codimension c , with its inherited stabilizers. Locally, this means an inclusion

$$[W/G] \hookrightarrow [U/G]$$

for a G -invariant complex submanifold $W \subset U$. For simplicity, assume $\mathcal{X}$ and $\mathcal{Y}$ are connected and compact. Write

$$Y = |\mathcal{Y}|, \quad r_{\mathcal{Y}} = |\text{generic stabilizer of } \mathcal{Y}|.$$

The weighted homology class in X is

$$\boxed{i_*[\mathcal{Y}]_{\text{orb}} = \frac{1}{r_{\mathcal{Y}}} i_*[Y]_{\text{top}}.} \quad (5.1)$$

This is the convention of [Z, Remark 1.21]. Define its **cohomological cycle class** by

$$\text{cl}_{\mathcal{X}}(\mathcal{Y}) := D_{\mathcal{X}}^{-1}(i_*[\mathcal{Y}]_{\text{orb}}) \in H^{2c}(X; \mathbb{Q}).$$

Its defining property is

$$\boxed{\int_{\mathcal{X}} \text{cl}_{\mathcal{X}}(\mathcal{Y}) \smile a = \int_{\mathcal{Y}} i^* a} \quad (5.2)$$

for every $a \in H^{d-2c}(X; \mathbb{Q})$.

Let PD_X^{top} denote ordinary rational Poincaré duality normalized using $[X]_{\text{top}}$. Since

$$D_{\mathcal{X}} = \frac{1}{r_{\mathcal{X}}} D_X^{\text{top}},$$

we obtain

$$\boxed{\text{cl}_{\mathcal{X}}(\mathcal{Y}) = \frac{r_{\mathcal{X}}}{r_{\mathcal{Y}}} \text{PD}_X^{\text{top}}(i_*[Y]_{\text{top}}).} \quad (5.3)$$

This is the normalization that is easiest to miss:

The coefficient in the homology cycle is $1/r_{\mathcal{Y}}$. The coefficient in its cohomological Poincaré dual, relative to the ordinary coarse-space dual, is $r_{\mathcal{X}}/r_{\mathcal{Y}}$.

As a consistency check, taking $\mathcal{Y} = \mathcal{X}$ gives

$$\text{cl}_{\mathcal{X}}(\mathcal{X}) = 1.$$

For an irreducible singular closed complex subvariety, one similarly uses its fundamental cycle, weighted by its generic stabilizer, and applies $D_{\mathcal{X}}^{-1}$. Smoothness of the subvariety is needed for the simple normal-bundle construction below, not for the existence of its cohomological cycle class. For a reducible cycle, the construction is additive, with the generic stabilizer taken separately on each component.

6. Differential-form representatives: Thom forms

6.1. Construction

For a smooth closed suborbifold $\mathcal{Y} \subset \mathcal{X}$ of complex codimension c , let

$$N = N_{\mathcal{Y}/\mathcal{X}}$$

be its complex normal orbifold bundle, of real rank $2c$.

Choose an orbifold tubular neighborhood and a closed Thom form

$$\tau \in \Omega^{2c}(N)$$

whose support is proper over $\mathcal{Y}$. Its normalization is that the integral over each **ordinary upstairs normal vector space** is 1.

Locally, write $\mathcal{Y} = [W/G] \subset [U/G] = \mathcal{X}$. An ordinary Thom form upstairs can be made invariant by averaging:

$$\tau_{\text{av}} = \frac{1}{|G|} \sum_{g \in G} g^* \tau.$$

The normal action preserves orientation, so averaging preserves the fiber integral 1. Compatible choices can be made using the usual bundle construction and invariant auxiliary data.

After shrinking its support into a tubular neighborhood and extending by zero, the resulting global closed form represents

$$[\tau] = \text{cl}_{\mathcal{X}}(\mathcal{Y}) \in H^{2c}(X; \mathbb{R}),$$

where the right-hand side denotes the real image of the rational cycle class.

6.2. Why there is no extra factor in the Thom form

For a compactly supported top-degree form β on W , ordinary fiber integration gives

$$\frac{1}{|G|} \int_{N_{W/U}} \tau \wedge p^* \beta = \frac{1}{|G|} \int_W \beta.$$

The same factor appears on both sides. Globally this is the defining identity

$$\int_{\mathcal{X}} [\tau] \smile a = \int_Y i^* a.$$

Therefore:

The ordinary upstairs fiber integral of the Thom form is 1. The orbifold integration rule supplies the stabilizer denominator; it should not be inserted into the Thom form a second time.

6.3. A point with stabilizer

Let $x \in X$ have stabilizer G_x . In a centered chart, choose a G_x -invariant top-degree bump form τ_x satisfying

$$\int_U \tau_x = 1.$$

Then

$$\int_{\mathcal{X}} \tau_x = \frac{1}{|G_x|}.$$

This form represents the dual of the weighted point cycle

$$[\{x\}]_{\text{orb}} = \frac{1}{|G_x|} [x]_{\text{top}}.$$

6.4. A caution about integration over chains

The chartwise integration rule should not be interpreted as redefining ordinary singular-cochain evaluation by attaching $1/|G_x|$ to every ordinary point simplex.

An ordinary coarse point represents $[x]_{\text{top}}$, and the constant cohomology class 1 evaluates on it as 1. The orbifold point BG_x represents the different rational cycle

$$\frac{1}{|G_x|} [x]_{\text{top}},$$

on which 1 evaluates as $1/|G_x|$.

For arbitrary chains, lifts and multiplicities must be handled compatibly. For integration and intersection theory, the weighted fundamental-cycle formulation makes the distinction unambiguous.

7. Intersection products, Chern classes, and pushforwards

The constructions above recover the standard intersection-theoretic formalism. See [B] for cycle classes and intersection products expressed through cohomology and duality.

7.1. Transverse intersections

Let $\mathcal{Y}, \mathcal{Z} \subset \mathcal{X}$ be closed complex suborbifolds meeting transversely in the upstairs charts. Then

$$\mathrm{cl}_{\mathcal{X}}(\mathcal{Y}) \smile \mathrm{cl}_{\mathcal{X}}(\mathcal{Z}) = \mathrm{cl}_{\mathcal{X}}(\mathcal{Y} \cap \mathcal{Z}).$$

One sees this by taking the product of Thom forms. If their dimensions are complementary, then

$$\boxed{\int_{\mathcal{X}} \mathrm{cl}_{\mathcal{X}}(\mathcal{Y}) \smile \mathrm{cl}_{\mathcal{X}}(\mathcal{Z}) = \sum_{p \in |\mathcal{Y} \cap \mathcal{Z}|} \frac{1}{|\mathrm{Aut}(p)|}.} \quad (7.1)$$

For proper isolated complex intersections that are not transverse, the corresponding local contribution is the upstairs intersection multiplicity divided by the stabilizer order. Self-intersections and excess intersections require the usual normal-bundle or excess-bundle classes, rather than a naive count of points.

7.2. Orbifold vector bundles and Chern classes

An orbifold vector bundle is given by equivariant bundles on charts and compatible gluing. Connections can be constructed by averaging locally and patching. Their curvature forms give global invariant Chern–Weil forms representing the real images of rational classes

$$c_j(E) \in H^{2j}(X; \mathbb{Q}).$$

Equivalently, the intrinsic Chern classes on $B\mathcal{X}$ can be transported to X using the rational cohomology comparison. Bundles and connections are introduced immediately after differential forms in [Z, §1.3].

For line bundles, rationality has a concrete explanation. On a compact orbifold, choose N divisible by the orders of the groups in a finite chart cover. Every stabilizer then acts trivially on the fibers of $L^{\otimes N}$. This power descends to a topological line bundle $\overline{L}_N$ on the coarse space, so

$$c_1(L) = \frac{1}{N} c_1(\bar{L}_N) \quad \text{in } H^2(X; \mathbb{Q}). \quad (7.2)$$

For a holomorphic orbifold line bundle and a nonzero meromorphic section s ,

$$c_1(L) = \text{cl}_{\mathcal{X}}(\text{div}(s)), \quad (7.3)$$

with zero and pole orders measured upstairs. This is the principle used in divisor computations of ψ -classes in $[Z]$.

For a complex vector bundle of higher rank, a transverse section, when one exists, gives the top Chern class as the cycle class of its zero locus. One should not assume without qualification that equivariant sections can always be made transverse.

7.3. Proper pushforwards

Let

$$f : \mathcal{X} \longrightarrow \mathcal{Z}$$

be a proper map between compact oriented smooth orbifolds, and let $|f| : X \rightarrow Z$ be its coarse map. Define

$$f_! = D_Z^{-1} \circ |f|_* \circ D_{\mathcal{X}}. \quad (7.4)$$

Its degree shift is

$$f_! : H^k(X; \mathbb{Q}) \longrightarrow H^{k + \dim_{\mathbb{R}} \mathcal{Z} - \dim_{\mathbb{R}} \mathcal{X}}(Z; \mathbb{Q}).$$

By construction,

$$\int_Z f_!(a) \smile b = \int_{\mathcal{X}} a \smile f^* b, \quad (7.5)$$

and the projection formula holds:

$$f_!(a \smile f^* b) = f_!(a) \smile b.$$

This is the duality-based pushforward formalism used in orbifold intersection theory.

Suppose the orbifolds are connected and of equal dimension, and the coarse map has degree δ . Then

$$\boxed{f_!(1) = \delta \frac{r_{\mathcal{Z}}}{r_{\mathcal{X}}}.} \quad (7.6)$$

Indeed,

$$|f|_*[\mathcal{X}]_{\text{orb}} = \frac{\delta}{r_{\mathcal{X}}} [Z]_{\text{top}} = \delta \frac{r_{\mathcal{Z}}}{r_{\mathcal{X}}} [\mathcal{Z}]_{\text{orb}}.$$

For orientation-preserving generically finite maps, δ is the usual generic coarse degree. More generally it is the signed topological degree.

In particular, the map $\text{pt} \rightarrow BG$ has orbifold degree $|G|$, whereas $BG \rightarrow \text{pt}$ has degree $1/|G|$, even though both coarse maps are maps from a point to a point. Here degree means the rational cohomological degree in (7.6); the definition does not require representability.

8. Examples fixing the normalizations

8.1. The effective quotient $[\mathbb{P}^1/\mu_m]$

Let μ_m act on $\mathbb{P}^1$ by rotations, fixing 0 and ∞ . Its generic stabilizer is trivial, and the coarse space is $\mathbb{P}^1$.

Let

$$h \in H^2(\mathbb{P}^1; \mathbb{Q}), \quad \langle h, [\mathbb{P}^1]_{\text{top}} \rangle = 1.$$

Then

$$\int_{\mathcal{X}} h = 1.$$

For an ordinary point p ,

$$\text{cl}_{\mathcal{X}}(p) = h.$$

For the stacky point at 0,

$$\text{cl}_{\mathcal{X}}(B\mu_m \text{ at } 0) = \frac{h}{m}, \quad \int_{\mathcal{X}} \frac{h}{m} = \frac{1}{m}.$$

There is no conflict with the global quotient rule. The atlas pullback of the coarse class h has degree m on the upstairs $\mathbb{P}^1$, and quotient integration divides that integral by m .

This is the phenomenon of [Z, Example 1.22]: the coarse points are homologous, but their orbifold-weighted point cycles are not equal.

8.2. The noneffective example $\overline{\mathcal{M}}_{1,1}$

The coarse space of $\overline{\mathcal{M}}_{1,1}$ is $\mathbb{P}^1$, and the generic stabilizer has order 2. The cusp also has stabilizer of order 2. These are the stabilizer normalizations used in the genus-one calculation in [Z, Proposition 2.21].

Let h again be the coarse class normalized by

$$\langle h, [\mathbb{P}^1]_{\text{top}} \rangle = 1.$$

Then

$$[\overline{\mathcal{M}}_{1,1}]_{\text{orb}} = \frac{1}{2}[\mathbb{P}^1]_{\text{top}}, \quad \int_{\overline{\mathcal{M}}_{1,1}} h = \frac{1}{2}.$$

Let $\mathcal{D}$ be the cusp. Its weighted **homology** class is

$$[\mathcal{D}]_{\text{orb}} = \frac{1}{2}[p]_{\text{top}}.$$

Its **cohomological** cycle class, however, is

$$\boxed{\text{cl}_{\overline{\mathcal{M}}_{1,1}}(\mathcal{D}) = \frac{2}{2}h = h,}$$

not $h/2$.

Let L_1 be the cotangent line bundle at the marked point, with $\psi_1 = c_1(L_1)$. The weight-12 discriminant gives a section of $L_1^{\otimes 12}$ with a simple zero at $\mathcal{D}$. Therefore

$$12\psi_1 = \text{cl}_{\overline{\mathcal{M}}_{1,1}}(\mathcal{D}) = h.$$

It follows that

$$\boxed{\psi_1 = \frac{h}{12}, \quad \int_{\overline{\mathcal{M}}_{1,1}} \psi_1 = \frac{1}{12} \cdot \frac{1}{2} = \frac{1}{24}.}$$

The factors have different meanings: $1/12$ belongs to the cohomology class relative to the coarse generator, while $1/2$ belongs to the ambient integration functional.

9. Normalization summary

Let $\mathcal{X}$ be connected, compact, and oriented, and let $\mathcal{Y} \subset \mathcal{X}$ be a connected closed oriented suborbifold with inherited stabilizers. With inclusion pushforwards understood, the three compatible normalizations are

$$\begin{aligned} [\mathcal{Y}]_{\text{orb}} &= \frac{1}{r_{\mathcal{Y}}} [Y]_{\text{top}}, \\ [\mathcal{X}]_{\text{orb}} &= \frac{1}{r_{\mathcal{X}}} [X]_{\text{top}}, \\ \text{cl}_{\mathcal{X}}(\mathcal{Y}) &= \frac{r_{\mathcal{X}}}{r_{\mathcal{Y}}} \text{PD}_X^{\text{top}}([Y]_{\text{top}}). \end{aligned}$$

They are characterized by the single compatibility identity

$$\int_{\mathcal{X}} \text{cl}_{\mathcal{X}}(\mathcal{Y}) \smile a = \int_{\mathcal{Y}} i^* a.$$

Locally, all these conventions come from the rule

$$\int_{[U/G]} \omega = \frac{1}{|G|} \int_U \tilde{\omega},$$

with the full group order, including the ineffective kernel.

References

[Z] Dimitri Zvonkine, *An Introduction to Moduli Spaces of Curves and Their Intersection Theory*. In particular: §1.3 for orbifolds, differential forms, integration, and bundles; Theorem 1.20 for rational Poincaré duality; Remark 1.21 for weighted fundamental cycles; Example 1.22 for weighted point classes; and Proposition 2.21 for the genus-one cotangent-line integral.

[B] Kai Behrend, *Cohomology of Stacks*. See the differential-geometric cohomology and intersection-theory constructions, and Proposition 36 for the comparison with coarse-space rational cohomology. Author's PDF: <https://personal.math.ubc.ca/~behrend/CohSta-1.pdf>.